

Quantifying the financial impact of supply chains

**How net income could be improved by saving 1% of COGS and the study of correlation
between net income and market cap**

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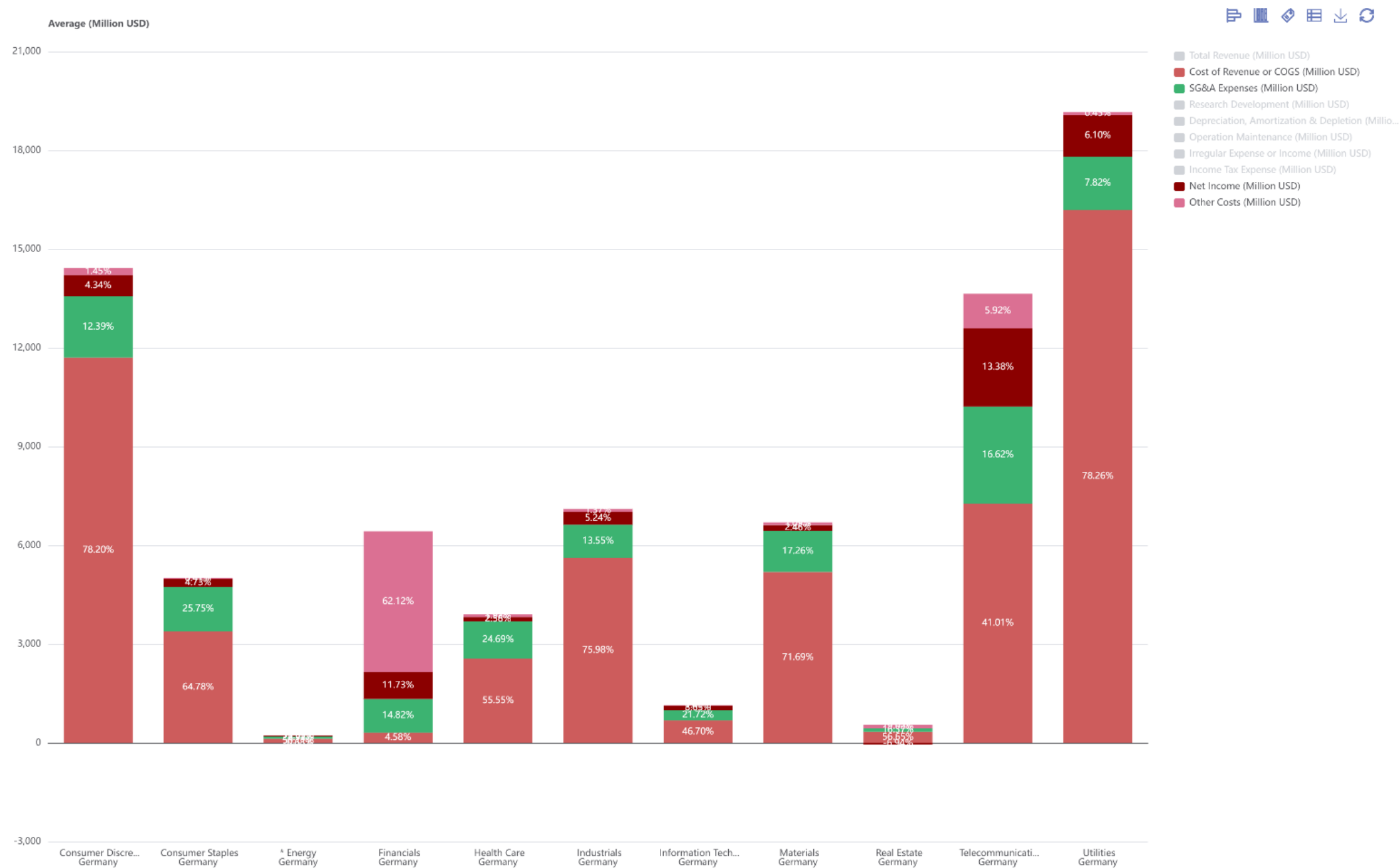
Supply Chain Analyst

Outlines

- Percentage of net margin improvement by saving 1% of COGS
- The study of correlation between market cap and net income
- The study of correlation between market cap and net income for each company

**Percentage of net margin improvement by saving
1% of COGS**

Industrial Breakdown of Germany - Revenue: Value of Operations / SCs.



* These industry and country combination(s) have a sample size smaller than 5.

7 out of 11 industry have COGS larger than 50 %. Thus 1% COGS lead to improvement in net margin. For example, consumer-Staples industry in Germany by calculating according to formula, 1% of COGS lead to 18% improvement in net margin.

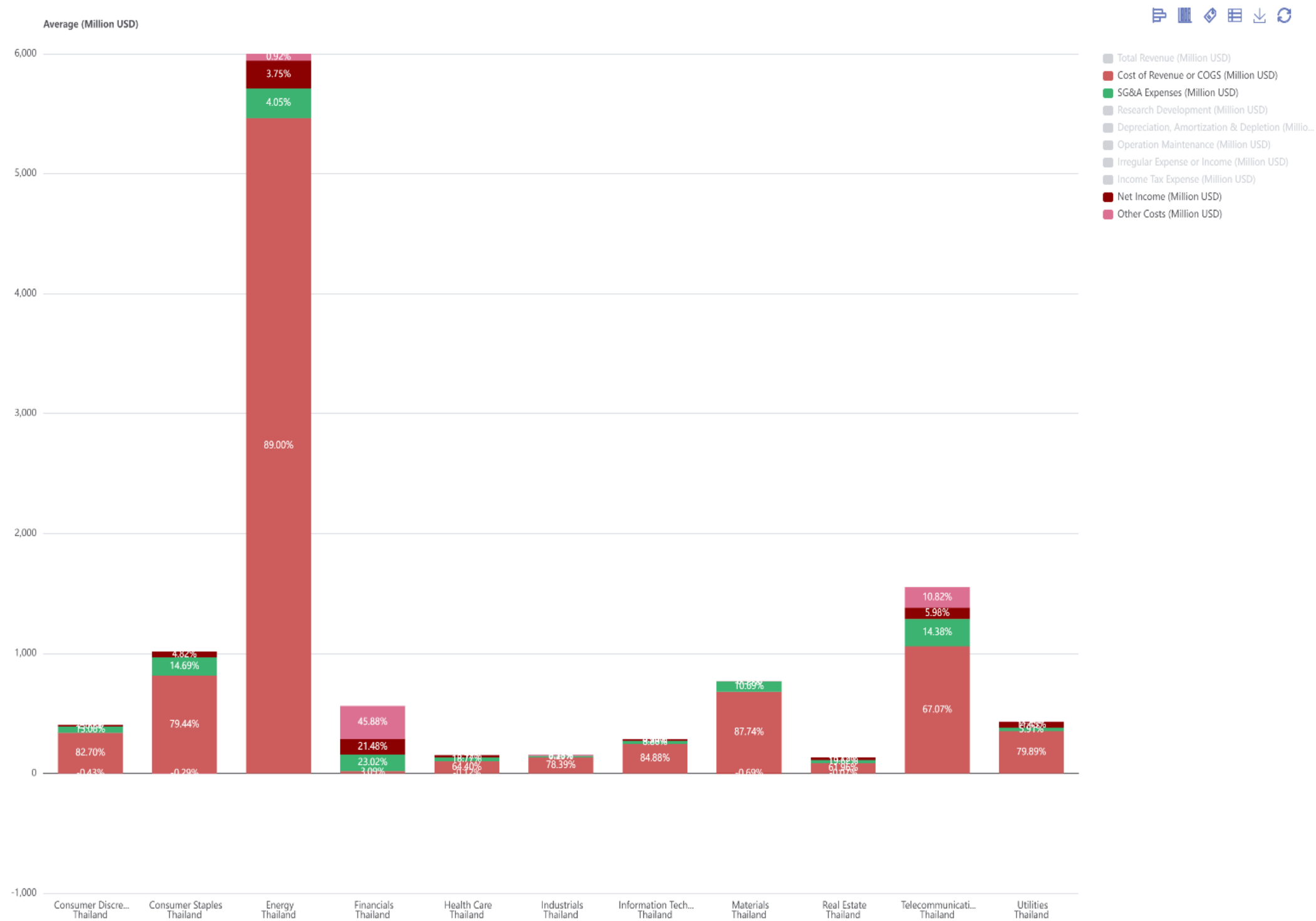
$$1 \% \text{ saving of COGS} = (\text{COGS} / \text{Revenue}) \times 1\%$$

$$= 0.782 \%$$

$$\% \text{ Increase in Net Income} = 1 \% \text{ saving of COGS} / (\text{Net Income} / \text{Revenue})$$

$$= 18\%$$

Industrial Breakdown (Thailand) - Revenue



10 out of 11 industry have COGS larger than 50 %. Thus 1% COGS lead to improvement in net margin. For example, Energy industry in Thailand, by calculating according to formula, 1% of COGS lead to 23% improvement in net margin.

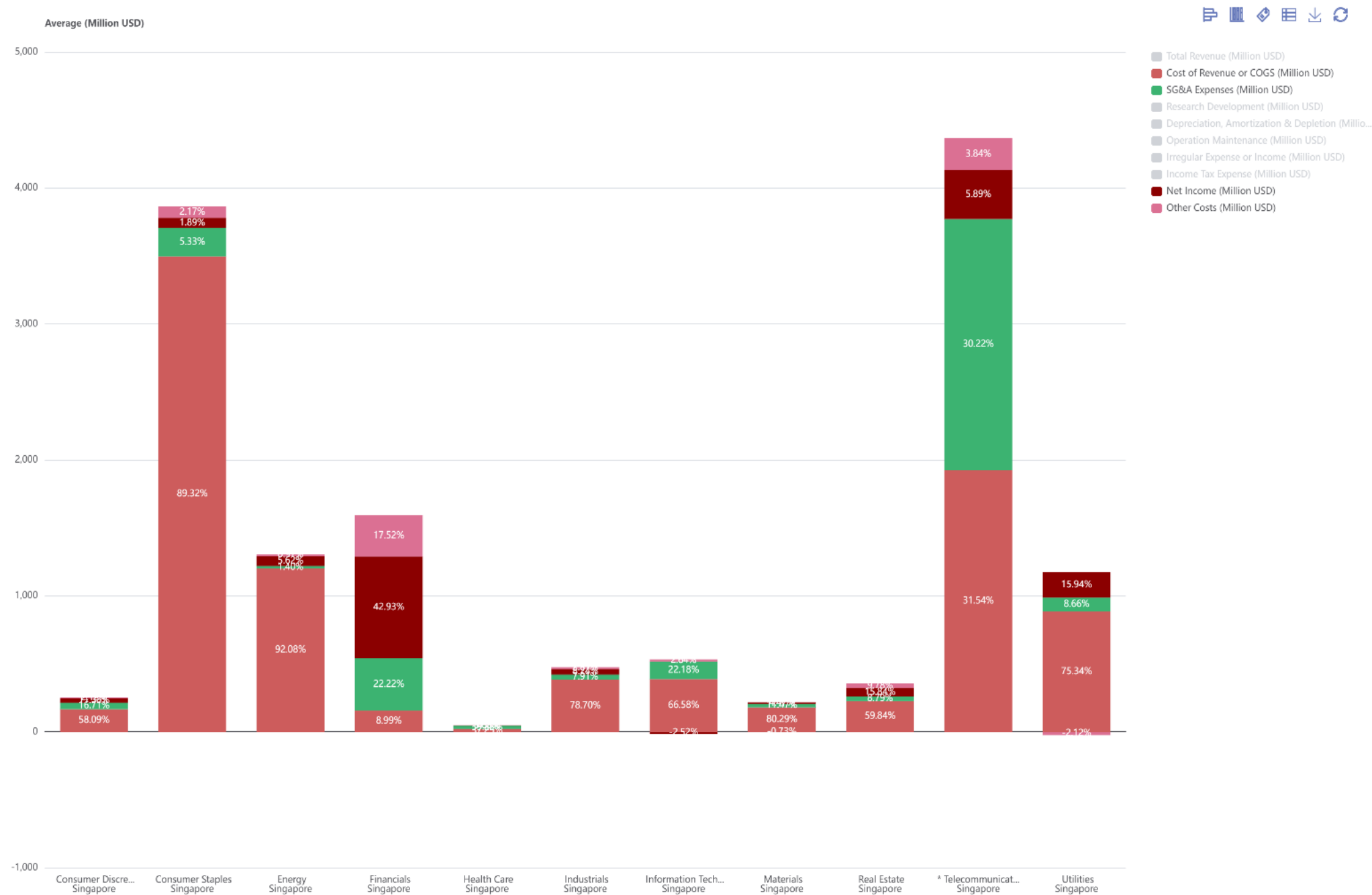
$$1 \% \text{ saving of COGS} = (\text{COGS} / \text{Revenue}) - 1\%$$

$$= 0.89 \%$$

$$\% \text{ Increase in Net Income} = 1 \% \text{ saving of COGS} / (\text{Net Income} / \text{Revenue})$$

$$= 23\%$$

Industrial Breakdown (Singapore) - Revenue



* These industry and country combination(s) have a sample size smaller than 5.

8 out of 11 industry have COGS larger than 50 %. Thus 1% COGS lead to improvement in net margin. For example, Energy industry in Singapore, by calculating according to formula, 1% of COGS lead to 16% improvement in net margin.

$$1 \% \text{ saving of COGS} = (\text{COGS} / \text{Revenue}) \times 1\%$$

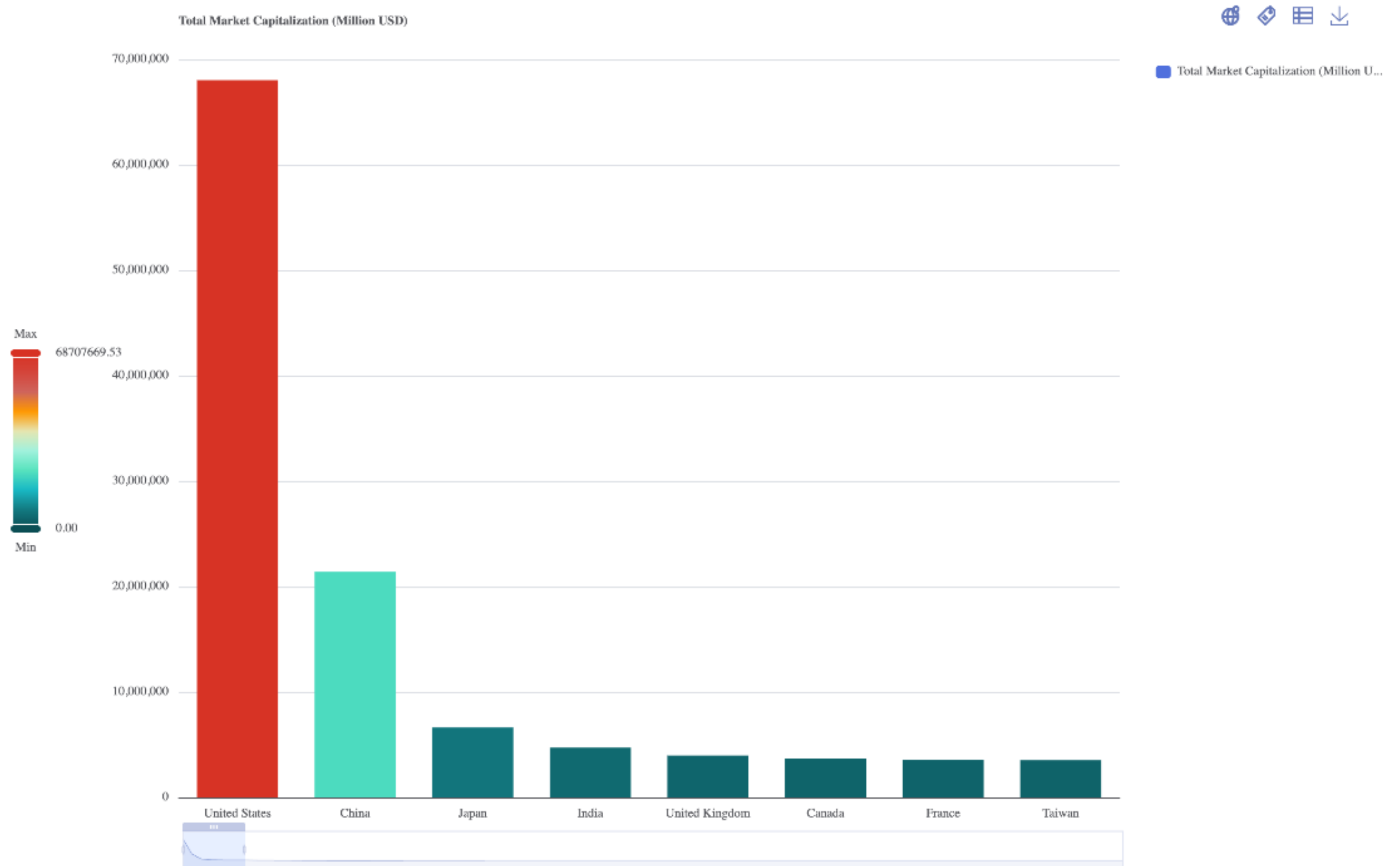
$$= 0.92 \%$$

$$\% \text{ Increase in Net Income} = 1 \% \text{ saving of COGS} / (\text{Net Income} / \text{Revenue})$$

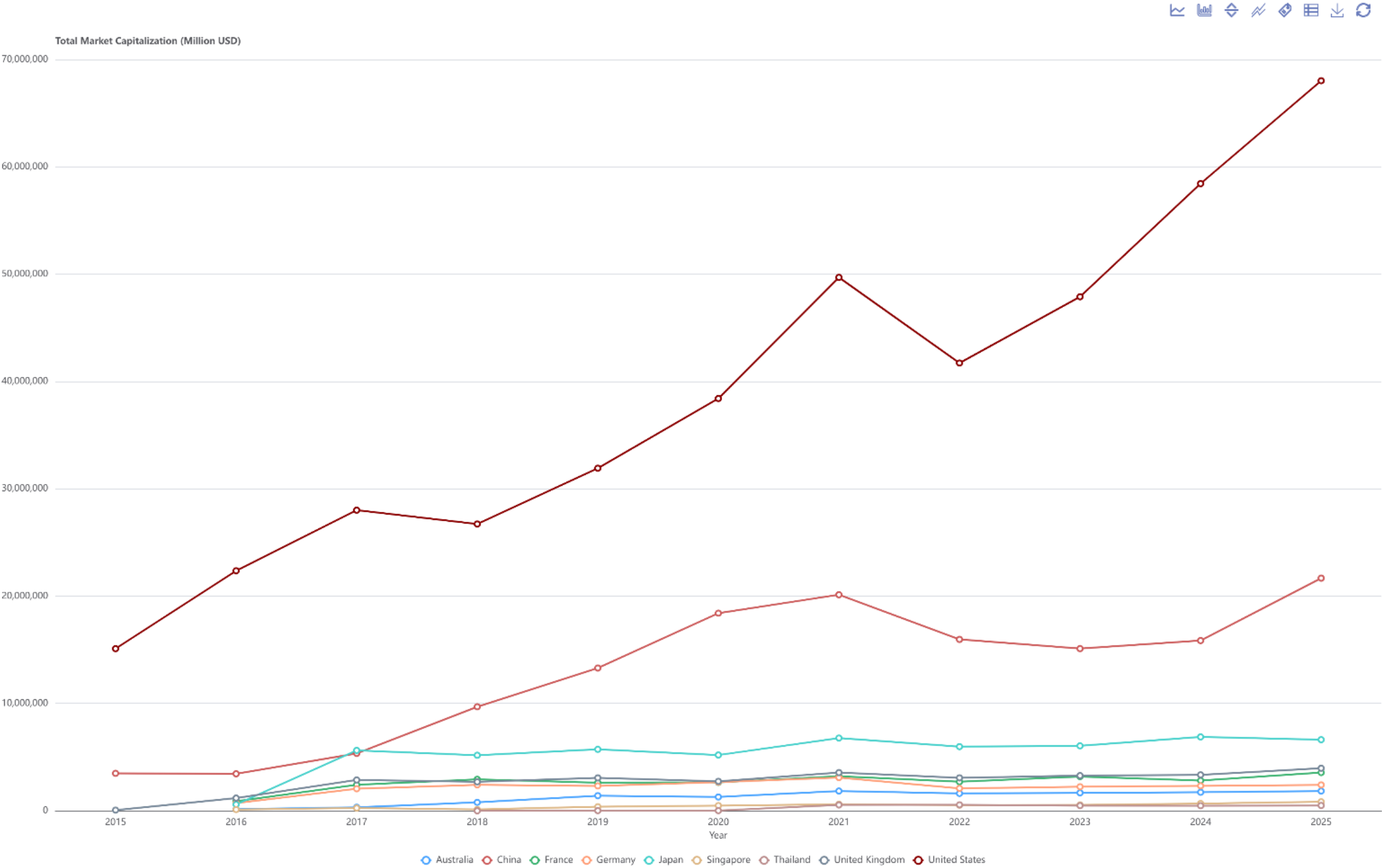
$$= 16\%$$

The Study of Correlation between market cap and net income

Total Market Capitalization

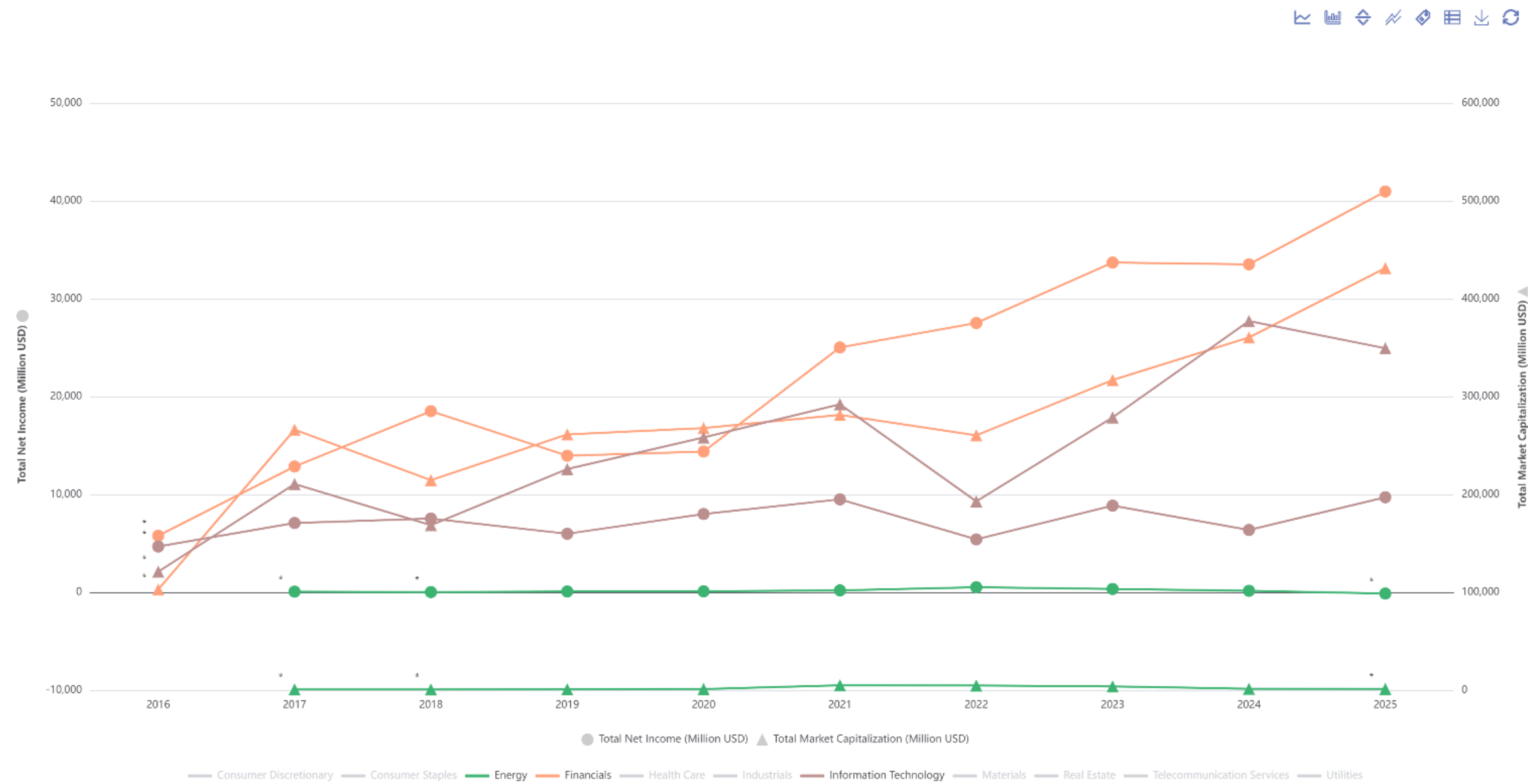


Total Market Cap Trend By Country 2025



United States is the highest market capitalization around the world.

Industrial Trend - Germany



Industrial market cap is correlated to net income. This graph represents some industry sectors from Germany.

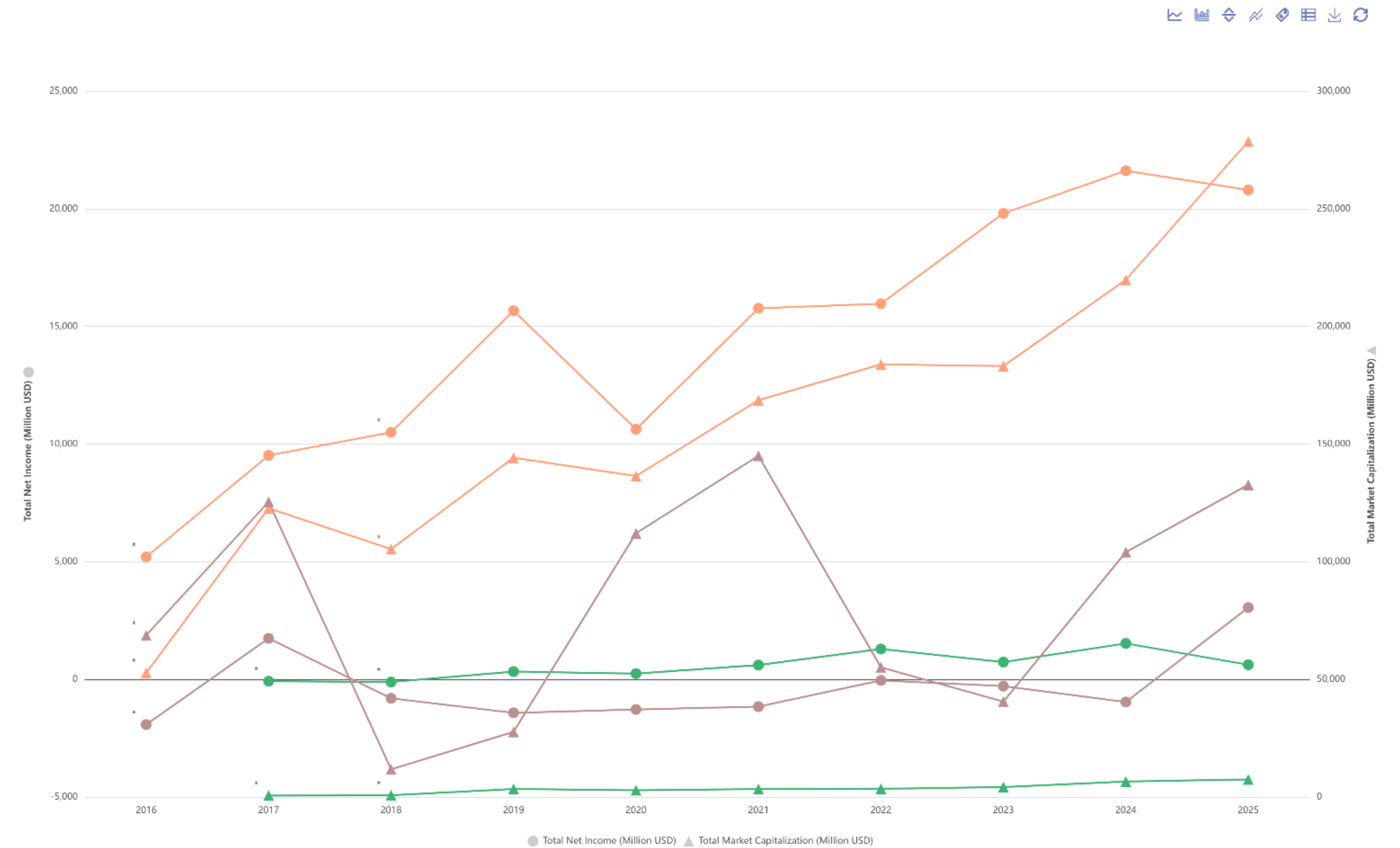
Industrial Trend - Thailand



* These KPI and Year combination(s) have a sample size smaller than 5.

Industrial market cap is correlated to net income. This graph represents some industry sectors from Thailand.

Industrial Trend - Singapore

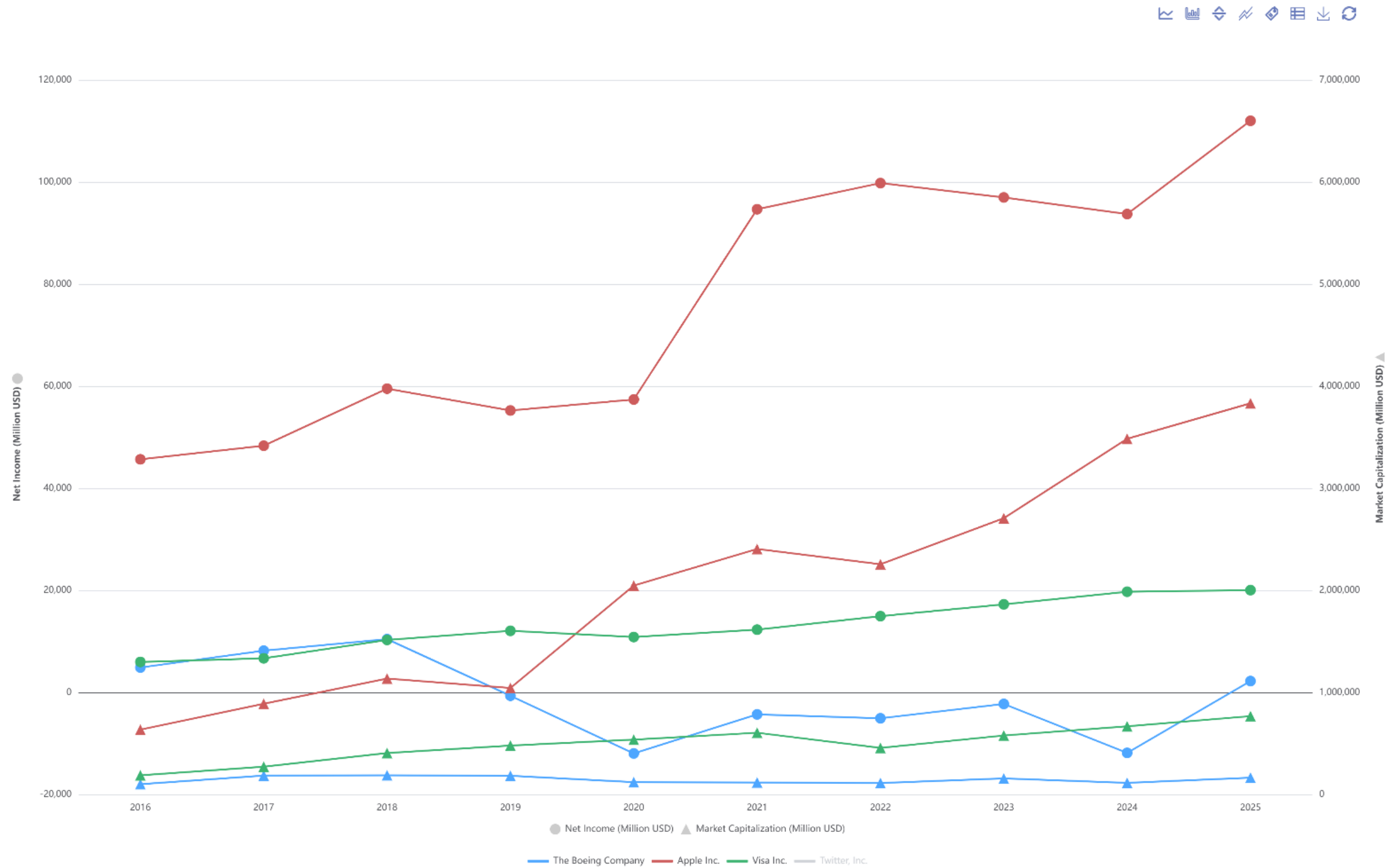


* These KPI and Year combination(s) have a sample size smaller than 5.

Industrial market cap is correlated to net income. This graph represents some industry sectors from Singapore.

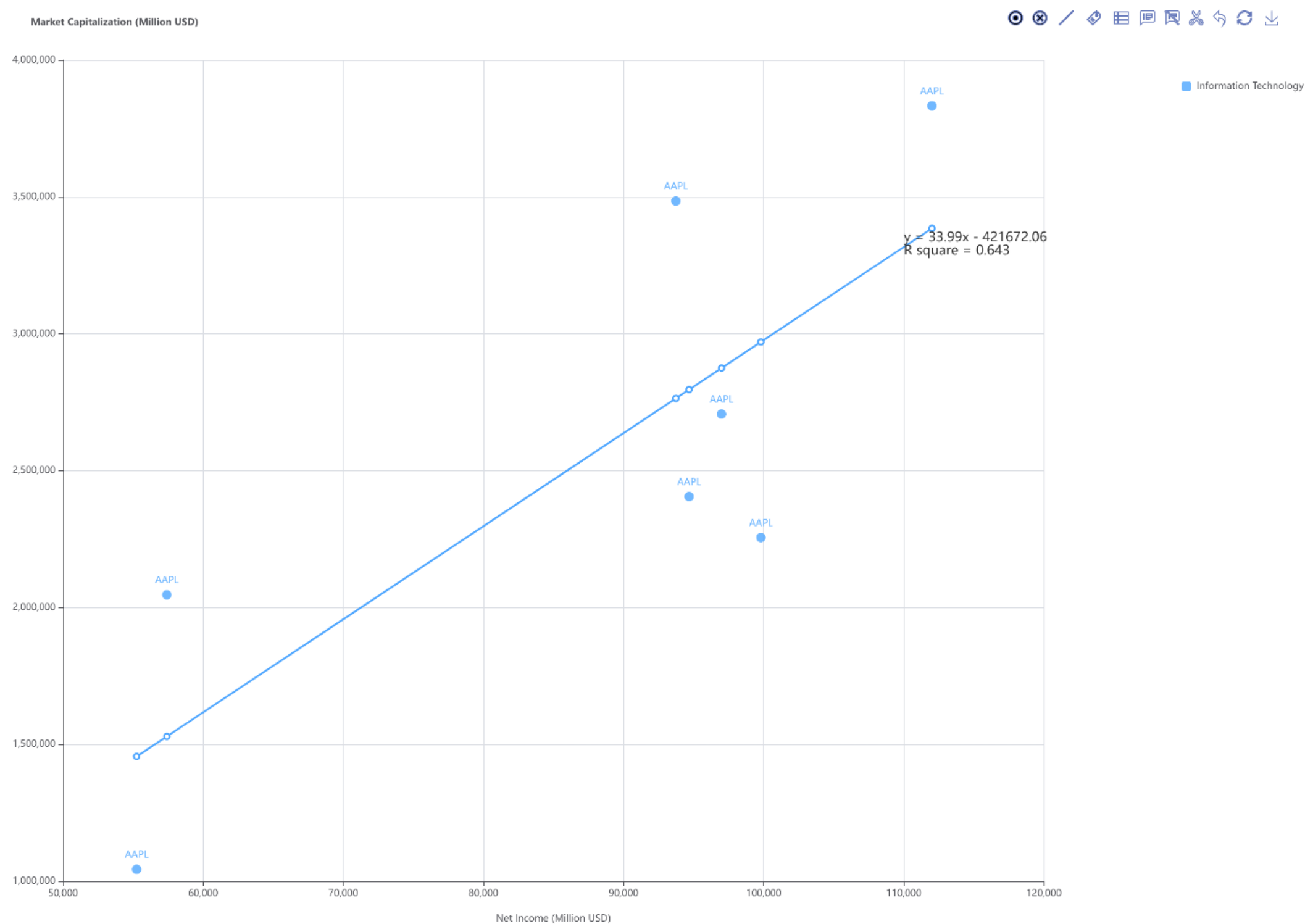
The Study of Correlation between market cap and net income of companies

Enterprise Trend

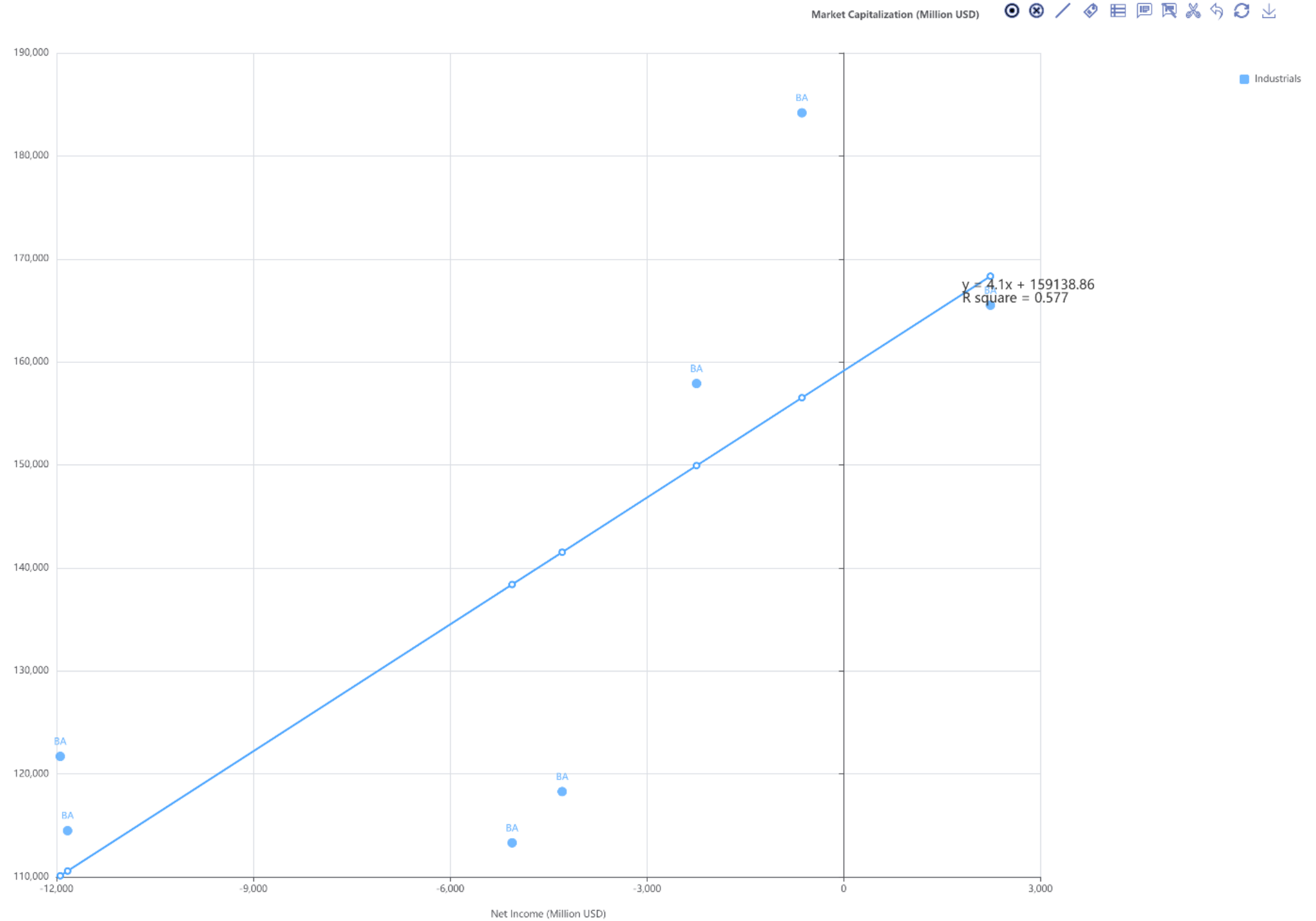


This graph illustrates the correlation between net income and market cap of 3 companies from US.

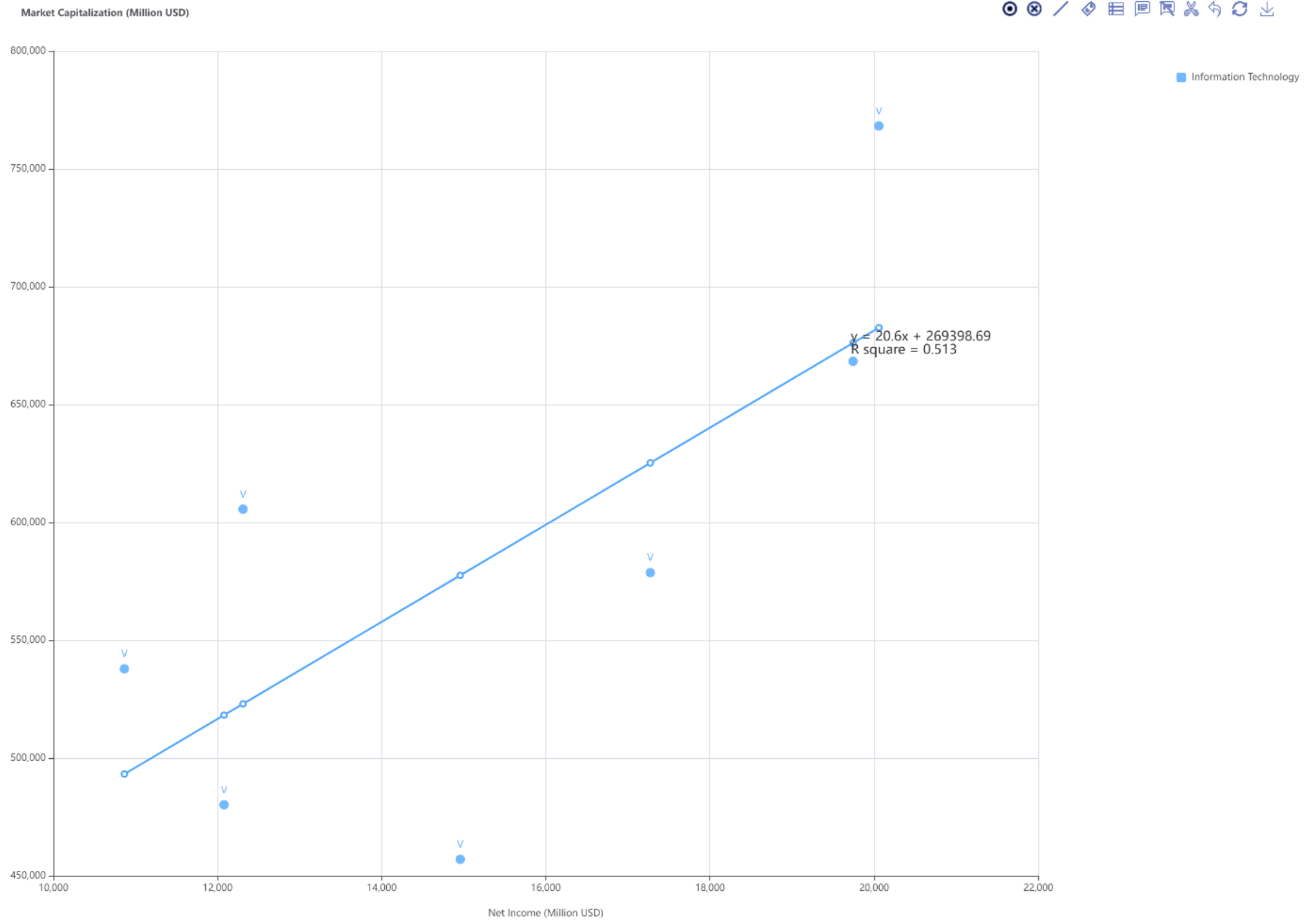
Apple Inc - \$1B change in net income --- > \$33.99 B change in market cap



The Boeing - \$1B change in net income --- > \$4.1 B change in market cap



Visa Inc - \$1B change in net income --- > \$20.6 B change in market cap



The Supply Chain Butterfly Effect

